

Selected robustness metrics

Continuous values: mean +/- sample SD; control roughness is mean per transition (lower is smoother); roughness/effort/steps: successful trials only

Case / variant	trials	Success	Control roughness overall (mean/transition)	Control roughness after wall (mean/transition)	Control effort	Number of steps	Average time per step (s)
condition=dynamic_wall variant=control_bank_mpp	400	313/400 (78.2%)	0.267 +/- 0.064 (n=313)	0.224 +/- 0.082 (n=313)	258.683 +/- 31.309 (n=313)	53.620 +/- 7.447 (n=313)	0.006 +/- 0.001 (n=400)
condition=dynamic_wall variant=corridor_prior_map	400	291/400 (72.8%)	0.312 +/- 0.042 (n=291)	0.347 +/- 0.048 (n=291)	192.795 +/- 30.303 (n=291)	90.876 +/- 9.864 (n=291)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=frenet_corridor_map	400	299/400 (74.8%)	0.277 +/- 0.037 (n=299)	0.294 +/- 0.042 (n=299)	185.634 +/- 30.181 (n=299)	90.826 +/- 9.274 (n=299)	0.012 +/- 0.003 (n=400)
condition=dynamic_wall variant=full_swarm_prior_map	400	302/400 (75.5%)	0.312 +/- 0.041 (n=302)	0.341 +/- 0.047 (n=302)	187.894 +/- 30.903 (n=302)	97.368 +/- 10.656 (n=302)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=gaussian_prior_map	400	302/400 (75.5%)	0.312 +/- 0.041 (n=302)	0.341 +/- 0.047 (n=302)	187.894 +/- 30.903 (n=302)	97.368 +/- 10.656 (n=302)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=heatmap_prior_map	400	291/400 (72.8%)	0.312 +/- 0.042 (n=291)	0.347 +/- 0.048 (n=291)	192.795 +/- 30.303 (n=291)	90.876 +/- 9.864 (n=291)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=mode_selecting_corridor_map	400	294/400 (73.5%)	0.381 +/- 0.041 (n=294)	0.424 +/- 0.049 (n=294)	233.699 +/- 33.371 (n=294)	72.177 +/- 11.883 (n=294)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=mode_selecting_homotopy_map	400	294/400 (73.5%)	0.381 +/- 0.041 (n=294)	0.424 +/- 0.049 (n=294)	233.699 +/- 33.371 (n=294)	72.177 +/- 11.883 (n=294)	0.009 +/- 0.003 (n=400)
condition=dynamic_wall variant=standard_mpp	400	48/400 (12.0%)	0.243 +/- 0.089 (n=48)	0.201 +/- 0.106 (n=48)	262.902 +/- 8.615 (n=48)	42.958 +/- 13.502 (n=48)	0.002 +/- 0.001 (n=400)
condition=no_wall variant=control_bank_map	100	96/100 (96.0%)	0.295 +/- 0.068 (n=96)	-	243.510 +/- 24.327 (n=96)	50.854 +/- 6.507 (n=96)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=corridor_prior_map	100	99/100 (99.0%)	0.314 +/- 0.038 (n=99)	-	176.210 +/- 23.455 (n=99)	85.384 +/- 8.068 (n=99)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=frenet_corridor_map	100	99/100 (99.0%)	0.271 +/- 0.037 (n=99)	-	169.745 +/- 24.654 (n=99)	86.465 +/- 8.396 (n=99)	0.005 +/- 0.001 (n=100)
condition=no_wall variant=full_swarm_prior_map	100	100/100 (100.0%)	0.306 +/- 0.035 (n=100)	-	171.892 +/- 26.595 (n=100)	93.610 +/- 10.594 (n=100)	0.003 +/- 0.001 (n=100)
condition=no_wall variant=gaussian_prior_map	100	100/100 (100.0%)	0.306 +/- 0.035 (n=100)	-	171.892 +/- 26.595 (n=100)	93.610 +/- 10.594 (n=100)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=heatmap_prior_map	100	99/100 (99.0%)	0.314 +/- 0.038 (n=99)	-	176.210 +/- 23.455 (n=99)	85.384 +/- 8.068 (n=99)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=mode_selecting_corridor_map	100	97/100 (97.0%)	0.375 +/- 0.042 (n=97)	-	219.067 +/- 24.006 (n=97)	67.639 +/- 11.125 (n=97)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=mode_selecting_homotopy_map	100	97/100 (97.0%)	0.375 +/- 0.042 (n=97)	-	219.067 +/- 24.006 (n=97)	67.639 +/- 11.125 (n=97)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=standard_mpp	100	29/100 (29.0%)	0.274 +/- 0.108 (n=29)	-	258.900 +/- 7.961 (n=29)	46.586 +/- 17.604 (n=29)	0.001 +/- 0.000 (n=100)
condition=static_wall variant=control_bank_map	400	397/400 (99.2%)	0.270 +/- 0.073 (n=397)	0.270 +/- 0.073 (n=397)	257.514 +/- 28.174 (n=397)	50.889 +/- 5.337 (n=397)	0.007 +/- 0.001 (n=400)
condition=static_wall variant=corridor_prior_map	400	391/400 (97.8%)	0.312 +/- 0.034 (n=391)	0.312 +/- 0.034 (n=391)	191.211 +/- 31.665 (n=391)	90.977 +/- 10.804 (n=391)	0.009 +/- 0.003 (n=400)
condition=static_wall variant=frenet_corridor_map	400	394/400 (98.5%)	0.274 +/- 0.034 (n=394)	0.274 +/- 0.034 (n=394)	187.318 +/- 28.954 (n=394)	92.061 +/- 9.805 (n=394)	0.010 +/- 0.003 (n=400)
condition=static_wall variant=full_swarm_prior_map	400	388/400 (97.0%)	0.305 +/- 0.038 (n=388)	0.305 +/- 0.038 (n=388)	187.944 +/- 29.232 (n=388)	96.546 +/- 10.234 (n=388)	0.008 +/- 0.003 (n=400)
condition=static_wall variant=gaussian_prior_map	400	388/400 (97.0%)	0.305 +/- 0.038 (n=388)	0.305 +/- 0.038 (n=388)	187.944 +/- 29.232 (n=388)	96.546 +/- 10.234 (n=388)	0.008 +/- 0.003 (n=400)
condition=static_wall variant=heatmap_prior_map	400	391/400 (97.8%)	0.312 +/- 0.034 (n=391)	0.312 +/- 0.034 (n=391)	191.211 +/- 31.665 (n=391)	90.977 +/- 10.804 (n=391)	0.009 +/- 0.003 (n=400)
condition=static_wall variant=mode_selecting_corridor_map	400	396/400 (99.0%)	0.374 +/- 0.045 (n=396)	0.374 +/- 0.045 (n=396)	228.739 +/- 30.866 (n=396)	69.306 +/- 9.322 (n=396)	0.011 +/- 0.004 (n=400)
condition=static_wall variant=mode_selecting_homotopy_map	400	396/400 (99.0%)	0.374 +/- 0.045 (n=396)	0.374 +/- 0.045 (n=396)	228.739 +/- 30.866 (n=396)	69.306 +/- 9.322 (n=396)	0.011 +/- 0.004 (n=400)
condition=static_wall variant=standard_mpp	400	46/400 (11.5%)	0.236 +/- 0.087 (n=46)	0.236 +/- 0.087 (n=46)	263.352 +/- 10.015 (n=46)	41.761 +/- 10.225 (n=46)	0.003 +/- 0.001 (n=400)

Failure reasons

Counts and percentages among failed trials

Case / variant	Failure reasons
condition=dynamic_wall variant=control_bank_mppi	not_reaching: 87 (100.0%)
condition=dynamic_wall variant=corridor_prior_mppi	not_reaching: 109 (100.0%)
condition=dynamic_wall variant=frenet_corridor_mppi	not_reaching: 101 (100.0%)
condition=dynamic_wall variant=full_swarm_prior_mppi	not_reaching: 98 (100.0%)
condition=dynamic_wall variant=gaussian_prior_mppi	not_reaching: 98 (100.0%)
condition=dynamic_wall variant=heatmap_prior_mppi	not_reaching: 109 (100.0%)
condition=dynamic_wall variant=mode_selecting_corridor_mppi	collision: 1 (0.9%); not_reaching: 105 (99.1%)
condition=dynamic_wall variant=mode_selecting_homotopy_mppi	collision: 1 (0.9%); not_reaching: 105 (99.1%)
condition=dynamic_wall variant=standard_mppi	collision: 1 (0.3%); not_reaching: 351 (99.7%)
condition=no_wall variant=control_bank_mppi	not_reaching: 4 (100.0%)
condition=no_wall variant=corridor_prior_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=frenet_corridor_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=full_swarm_prior_mppi	none
condition=no_wall variant=gaussian_prior_mppi	none
condition=no_wall variant=heatmap_prior_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=mode_selecting_corridor_mppi	not_reaching: 3 (100.0%)
condition=no_wall variant=mode_selecting_homotopy_mppi	not_reaching: 3 (100.0%)
condition=no_wall variant=standard_mppi	not_reaching: 71 (100.0%)
condition=static_wall variant=control_bank_mppi	not_reaching: 3 (100.0%)
condition=static_wall variant=corridor_prior_mppi	not_reaching: 9 (100.0%)
condition=static_wall variant=frenet_corridor_mppi	not_reaching: 6 (100.0%)
condition=static_wall variant=full_swarm_prior_mppi	not_reaching: 12 (100.0%)
condition=static_wall variant=gaussian_prior_mppi	not_reaching: 12 (100.0%)
condition=static_wall variant=heatmap_prior_mppi	not_reaching: 9 (100.0%)
condition=static_wall variant=mode_selecting_corridor_mppi	not_reaching: 4 (100.0%)
condition=static_wall variant=mode_selecting_homotopy_mppi	not_reaching: 4 (100.0%)
condition=static_wall variant=standard_mppi	collision: 1 (0.3%); not_reaching: 353 (99.7%)